

CIVE 202

Spring 2026

Project 2

Scope of Work

By Group 6:

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Submitted to: Civil and Environmental Engineering Department
faculty

Project Goals

The goal of this project is to recreate the Nebraska Department of Transportation's (NDOT) concrete mix design model, transforming it from an Excel-based to a Python-based system. By doing so, the client hopes to make the concrete mix calculations more transparent and repeatable. NDOT plans to use this tool to improve computation reliability and for possible future integration into a web-based tool. This model will intake material weights, properties, and ratios, and return the metrics corresponding to one cubic yard of the material described.

Project Tasks

Task #1 - Translate the Excel Logic into Python

The calculations from the Excel worksheet will be replicated in the python file, and global constants like unit water weight and ft^2 per yard will be defined. We will define specific functions for each calculation and place them sequentially such that the calculation will complete. This includes finding the volume of each cementitious material and aggregate weights. No outside research will be necessary for this step. We will use previously learned skills like function definition, constant referencing, and mathematical variable operations to complete this task.

Task #2 - Implement Sequential User Inputs

We will write prompts for the user to complete, asking the user to enter information about the concrete mix components. These will print one at a time, streamlining the data entry process compared to the Excel sheet. It will collect information on material weights, specific gravities, the water/cement ratio, and desired air content. No additional research will be necessary, and we will use new 'input' function to complete this task

Task #3 – Generate a Final Mix Design Output

In order to properly replicate the Excel data sheet, the calculations will need to be output in a clear and readable format. We will code a chart that displays the results of the process by using the inputs collected by Task #2 and running the variables through the appropriate functions from Task#1. Components like f-string formatting will be employed.

Task #4 – Prepare and Evaluate Four Concrete Mix Design Scenarios

To test the effectiveness of this tool, we will research and find 4 concrete mix designs to run through the model. This task will require research through both NDOT and outside resources. These scenarios must have the appropriate components in order to be input into the Python model and will verify the effectiveness of this tool.

Deliverables

After the completion of these tasks, the following items will be sent to the client NDOT.

Deliverable	Files Included	Due to Client by
Excel-Based Model	CIVE202_Spring2025_Group6_Project2_ExcelMixDesign.xlsx	2/24/26
Gantt Chart	CIVE202_Spring2025_Group6_Project2_GanttChart.xlsx	2/24/26
Python Code	CIVE202_Spring2025_Group6_Project2_ScopeOfWork.docx	2/24/26
Annotated Code Document	CIVE202_Spring2026_Group6_Project2_AnnotatedCodeDocument.docx	2/24/26
Scope of Work	CIVE202_Spring2025_Group6_Project2_ScopeOfWork.docx	2/24/26
Mix Research	CIVE202_Spring2026_Group6_Project2_MixDesignResearch.docx	2/24/26
Written Report	CIVE202_Spring2026_Group6_Project2_WrittenReport.docx	